More to know

DNA finger printing

Traditionally, one of the most accurate methods for placing an individual at the scene of a crime has been a fingerprint. With the advent of recombinant DNA technology, a more powerful tool is now available: DNA fingerprinting is (also called DNA typing or DNA profiling). It was first invented by Professor Sir Alec Jeffreysin 1984. The DNA finger print is unique for every person and can be extracted from traces of samples from blood, saliva, hair etc…By using this method we can detect the individual specific variation in human DNA.

In this method, the extracted DNA is cut at specific points along the strand with restriction of enzymes resulting in the formation of DNA fragments of varying lengths which were analysed by technique called gel electrophoresis. This method separates the fragments based on their size. The gel containing the DNA fragments is then transferred to a nylon sheet using a technique called blotting. Then, the fragments will undergo autoradiography in which they were exposed to DNA probes (pieces of synthetic DNA that were made radioactive and that bound to the fragments). A piece of X-ray film was then exposed to the fragments, and a dark mark was produced at any point where a radioactive probe had become attached. The resultant pattern of marks could then be compared with other samples. DNA fingerprinting is based on slight sequence differences (usually single base-pair changes) between individuals. These methods are proving decisive in court cases worldwide.

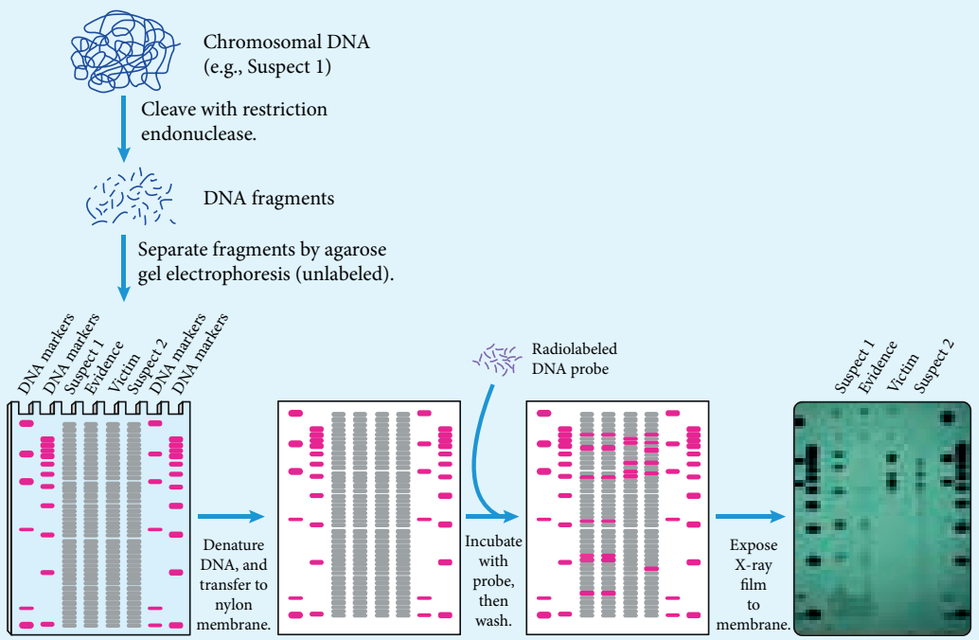


Figure 14.22 DNA finger printing

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